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Feasibility Study

**AR based Astronomy learning module for grade 08
students in Sri Lanka**

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1. Executive Summary

This feasibility study examines the use of Augmented Reality (AR) technology to enhance the teaching and learning of Astronomy for Grade 8 students in Sri Lanka. Astronomy topics such as planetary motion, inner planets, outer planets, and the Sun–Earth–Moon system are inherently spatial and dynamic, yet they are traditionally taught through static textbook diagrams and lecture-based explanations. These limitations make it difficult for students to visualize celestial processes, leading to common misconceptions and reduced conceptual understanding. The objective of this study is to develop a mobile AR application that enables students to visualize and interact with curriculum-aligned astronomy content through immersive 3D simulations displayed directly on their textbooks.

The proposed AR solution allows students to scan printed diagrams in the Grade 8 science textbook using a smartphone camera, triggering realistic 3D astronomy simulations developed with Unity and Vuforia’s image-target recognition. By overlaying dynamic celestial models on static diagrams, the system promotes inquiry-based learning and increases engagement. This approach supports both conceptual clarity and motivation without requiring high-cost laboratory equipment, telescopes, or continuous internet connectivity—making it suitable for typical Sri Lankan classrooms.

A feasibility evaluation was conducted to assess technical viability, operational suitability, market demand, financial considerations, and ethical alignment. Technical feasibility analysis indicates that Unity and Vuforia can effectively render astronomy simulations on low-cost Android devices while supporting offline operation. Market and educational feasibility studies point to an increasing interest in digital tools that enhance STEM learning and address visualization challenges in astronomy education. Additionally, teachers have reported the need for interactive tools that align with the national curriculum and reduce reliance on rote memorization.

The positive outcomes from this feasibility assessment support the development and deployment of the proposed AR-based astronomy application. As an accessible and scalable educational tool, it has the potential to significantly improve students’ conceptual understanding of celestial phenomena and increase interest in science learning. The proposed system aligns with modern pedagogical trends while addressing constraints in Sri Lankan schools, making it a viable and impactful solution for transforming astronomy education.

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3. Introduction

Background

Astronomy education introduces learners to celestial systems, planetary motion, and large-scale spatial relationships that are inherently dynamic and three-dimensional. For Grade 8 students in Sri Lanka, these concepts are typically taught through textbooks, diagrams, and teacher-led explanations. While these resources provide theoretical coverage, they fall short in supporting visualization and spatial reasoning. Students commonly struggle to understand phenomena such as lunar phases, eclipses, rotations, revolutions, and the relative scale of the solar system due to their abstract and non-observable nature. As a result, misconceptions frequently emerge, and student engagement declines.

Traditional classroom environments provide limited access to interactive tools or practical demonstrations. Schools often lack telescopes, digital visualization equipment, or simulation resources that could support deeper understanding. Lesson delivery largely relies on board-based instruction and memorization-based assessments, leading to surface-level learning rather than conceptual mastery. Teachers have also noted challenges in demonstrating astronomy content within the time and resource constraints of the standard curriculum.

Augmented Reality (AR) offers a viable solution to these challenges by enabling real-time visualization of celestial processes through mobile devices. By scanning textbook diagrams, students can view curriculum-aligned 3D astronomy simulations that illustrate motion, scale, and interactions from multiple perspectives. AR enhances conceptual understanding through active exploration, while also increasing motivation and curiosity in science learning. Its ability to operate on low-cost smartphones and in offline environments makes it suitable for Sri Lankan classrooms where digital infrastructure is limited.

The integration of AR into astronomy education represents a practical and scalable approach to modernizing STEM instruction, improving visualization, and reducing dependence on rote learning. This feasibility study evaluates the applicability of AR-based astronomy instruction in the Sri Lankan school system and assesses its potential to enhance learning outcomes for Grade 8 students.

Objectives

The study aims to:

- Assess the technological requirements for developing an interactive AR application to support astronomy education for Grade 8 students.
- Identify the market demand, target user groups, and adoption feasibility of AR-based astronomy learning tools within Sri Lankan schools.
- Analyze financial considerations, cost implications, and long-term sustainability of implementing AR for astronomy education.
- Address operational, ethical, and accessibility concerns associated with using AR in classroom environments.
- Evaluate the educational effectiveness of AR in improving conceptual understanding, visualization, and engagement in astronomy learning.

Scope

- **Technical Feasibility:** Evaluation of required hardware, software, and technical competencies for designing the AR-based astronomy learning application using Unity and Vuforia, including device compatibility and offline functionality.
- **Market Feasibility:** Identification of target users (Grade 8 students and geography teachers), assessment of demand for AR-supported astronomy learning, and comparison with existing educational resources and tools used in Sri Lankan schools.
- **Financial Feasibility:** Analysis of development costs, potential funding avenues, and long-term strategies for affordable deployment and maintenance of AR-based astronomy learning tools in schools.
- **Operational Feasibility:** Examination of classroom usability, integration with the national astronomy curriculum, teacher support needs, scalability across multiple school settings, and acceptance by stakeholders.
- **Environmental and Ethical Feasibility:** Consideration of accessibility, equitable use in resource-limited schools, safety and privacy concerns, and alignment with educational and ethical standards applicable to Sri Lankan students.

4. Problem Statement

Astronomy education in Sri Lankan schools relies primarily on textbooks, diagrams, and lecture-based instruction. While sufficient for theoretical delivery, these traditional methods struggle to convey the spatial, dynamic, and three-dimensional nature of astronomical phenomena. Grade 8 astronomy topics—such as planetary motion, lunar phases, eclipses, rotations and revolutions, and the relative scale of celestial bodies—require visualization and temporal reasoning that static diagrams cannot adequately provide. As a result, students often form misconceptions and experience difficulty in understanding how celestial systems behave and interact.

Current instructional practices do not offer interactive or inquiry-based learning opportunities, limiting students' ability to explore, manipulate, or observe astronomical processes in real time. Learning becomes passive, engagement declines, and conceptual understanding remains surface-level. Teachers also face challenges in demonstrating astronomy concepts within the time, resource, and equipment constraints of the standard classroom environment.

There is a clear need for an educational solution that enhances visualization, encourages active exploration, and bridges the gap between abstract scientific theory and student understanding. Augmented Reality (AR) presents a viable pathway by enabling mobile devices to overlay interactive 3D astronomy simulations directly on top of printed textbook content. Through real-time visualization and safe, accessible interaction, AR has the potential to improve student engagement, correct misconceptions, and strengthen conceptual understanding in astronomy education for Sri Lankan Grade 8 learners.

5. Project Description

The proposed AR-based astronomy learning system provides Grade 8 students with real-time, three-dimensional simulations of celestial processes to support deeper conceptual understanding. Using mobile devices, students can scan textbook diagrams to visualize and interact with 3D models representing planetary motion, lunar phases, eclipses, rotations and revolutions, and the relative scale of celestial bodies. These simulations allow learners to explore dynamic astronomical phenomena from multiple perspectives, observe changes over time, and understand spatial relationships that are otherwise difficult to convey through static textbooks and lectures.

The system utilizes Unity for 3D content development and Vuforia image-target recognition to overlay AR simulations directly on top of printed textbook content. Students can manipulate parameters, adjust viewpoints, and track temporal changes in the simulation environment, enabling active learning and reducing common misconceptions related to astronomy topics. All AR modules are aligned with the Sri Lankan Grade 8 science curriculum to ensure relevance and classroom applicability.

By combining visualization, interactivity, and curriculum alignment, the AR-based astronomy platform offers a low-cost, accessible, and engaging approach to enhancing conceptual understanding. The system supports offline operation and functions on budget-friendly Android devices, making it suitable for resource-constrained school environments. The proposed platform has the potential to improve student engagement, correct misconceptions, and strengthen teacher-led instruction, contributing to more effective astronomy education in Sri Lankan classrooms.

6. Feasibility Analysis

a. Technical Feasibility

Technical Requirements: Unity for AR development, Blender for 3D celestial models, Vuforia for image-target recognition, running on Android smartphones with offline support for typical Sri Lankan classrooms.

Development Stages: Plan astronomy content, create 3D models, integrate AR features (rotation, animation, time progression), test, and deploy.

Potential Challenges: Smooth real-time rendering on mid-range devices, maintaining scientific accuracy, avoiding misconceptions, and optimizing 3D assets.

Prototypes & Testing: AR prototypes tested with Grade 8 students and teachers to refine usability, animation, and curriculum alignment.

b. Market Feasibility

Target Audience: Grade 8 students and geography teachers in Sri Lankan schools looking for improved astronomy learning experiences.

Market Needs & Trends: Difficulty visualizing planetary motion, lunar phases, eclipses, and the solar system drives demand for interactive, curriculum-aligned AR tools.

Competitive Analysis: Most current apps are 2D. AR-based 3D simulations over textbooks provide a unique, hands-on experience that allows real-time exploration of celestial phenomena.

Evidence of Demand: Classroom trials and teacher feedback show strong interest in AR astronomy tools for better engagement and understanding.

c. Financial Feasibility

Cost Estimates: Estimated cost is Rs. 20,000–30,000, covering AR app development, creation of 3D celestial models, and necessary hardware/software (Android devices, Unity, Blender, Vuforia licenses).

d. Operational Feasibility

Implementation: Requires AR developers, UX designers, and 3D modelers for creating celestial simulations and app features.

Logistics & Scalability: AR modules can be tested and deployed in urban and rural schools.

Risks & Mitigation: Regular updates and feedback from teachers/students ensure usability, engagement, and long-term effectiveness.

e. Environmental or Ethical Feasibility

Environmental Impact: Fully digital AR simulations require no physical labs, posing minimal environmental risks.

Sustainability: Reduces need for resource-intensive teaching aids, supporting eco-friendly, low-cost STEM learning.

Ethical Considerations: Content is validated by teachers for accuracy, curriculum alignment, inclusivity, and student privacy.

7. Risk Analysis

Identified Potential Risks

1. Technical Risks

- **Device Compatibility:** AR app must run smoothly on mid-range Android devices commonly used in Sri Lankan schools.
- **Software Stability and Performance:** Ensure smooth real-time rendering of 3D celestial models and interactive simulations.
- **Content Accuracy:** Astronomy simulations (planetary motion, eclipses, lunar phases) must be scientifically accurate to avoid misconceptions.

2. Financial Risks

- **Development Costs:** Initial investment for AR app, 3D celestial models, and related software/hardware.
- **Adoption Challenges:** Schools may hesitate to adopt new AR tools due to budget or lack of familiarity.

3. Operational Risks

- **User Training and Adaptation:** Teachers and students may need guidance to use AR features effectively.
- **Technical Support and Maintenance:** Regular updates and troubleshooting are required for smooth classroom use.
- **Scalability Challenges:** Expanding AR modules to additional astronomy topics or schools may require extra resources and expertise.

Likelihood and Potential Impact

- **Technical Risks:** High impact if the AR app malfunctions or mid-range devices cannot render 3D celestial simulations smoothly.
- **Financial Risks:** Medium to high impact; limited school budgets or slow adoption may delay deployment.
- **Operational Risks:** Medium likelihood and medium impact, as teachers and students may need guidance to use AR effectively.

Mitigation Strategies

- **Technical Mitigation:**
 - Conduct extensive testing to ensure smooth AR performance on common Android devices.
 - Collaborate with geography teachers to validate the accuracy and curriculum alignment of astronomy simulations.
- **Financial Mitigation:**
 - Seek partnerships or grants from schools, educational organizations, or government programs.
 - Explore cost-effective sustainability models, such as school licenses or shared resources.
- **Operational Mitigation:**
 - Provide teacher training and student onboarding to ensure effective AR use.
 - Design a scalable framework to allow future expansion of astronomy modules and other STEM topics.

8. Recommendations

The development of the AR-based Grade 8 Astronomy learning system is strongly recommended based on the feasibility analysis. Early collaboration with teachers, curriculum experts, and students is essential to ensure accurate, curriculum-aligned content. A prototype phase should be conducted to test AR interactions, gather classroom feedback, and refine the 3D celestial simulations before full-scale deployment in schools.

9. Conclusion

The proposed AR-based Grade 8 Astronomy learning system offers a feasible and innovative approach to improving astronomy education in Sri Lankan classrooms. By using augmented reality, students can visualize and interact with complex celestial concepts—such as planetary motion, inner planets, and outer planets and eclipses—enhancing understanding and engagement. This solution has the potential to transform traditional textbook-based astronomy instruction and modernize STEM learning through immersive, curriculum-aligned experiences.

10. References

- [1] T. Gumilar, R. F. R. Uulaa, and S. Chen, “Exploring augmented reality on astronomy education: impacts on conceptual knowledge, motivation, and learning attitude,” *Proceedings of International Conference on Education*, 2023.
- [2] C. Chiang, S. Yang, and T. Hwang, “An augmented reality–based mobile learning system to improve students’ learning achievements and motivations in natural science inquiry,” *Educational Technology & Society*, vol. 17, no. 4, pp. 133–149, 2014. (background on AR in STEM)
- [3] “Creative situated augmented reality learning for astronomy curricula,” *Educational Technology & Society* (Cosmos Planet Go App AR study showing effectiveness).
- [4] A. Ramadhani, I. Ismet, and K. Wiyono, “Solar system learning innovation through augmented reality: increasing student concept mastery,” *Integrated Science Education Journal*, 202x.
- [5] “Enhancing conceptual understanding of the solar system through 3D augmented reality in primary education,” *Journal of Computing and Smart Ecosystems*.
- [6] M. A. Putra, M. Madlazim, and E. Hariyono, “Exploring AR-based learning media implementation in solar system materials,” *International Journal of Recent Educational Research*, 202x.
- [7] “Implementation of augmented reality applications in developing flashcard learning media for the solar system,” *Applied Sciences (MDPI)* — AR marker-based astronomy learning research.
- [8] Esra B. Özdemir, “Investigation of the effects of augmented reality applications on the astronomy literacy levels of secondary school students,” *TAY Journal*, vol. 7, no. 4 (2023).

12. Appendices

